

ONE DATE, TWO FUTURES

What if the 40 unburnt fields in my Jalandhar-area sample faced different residue-management conditions during the two weeks after 29 October 2021? I want to see how those assumptions change the number and location of fields that could burn.

125

FIELDS IN CLUSTER

40

UNBURNT / ELIGIBLE

85

ALREADY BURNT

THE DATASET / THE RECORD

CEEW, Patterns of Crop Residue Burning. Researchers traced fields and classified burn condition from very-high-resolution imagery observed on 29 Oct 2021. Released in 2026, the GeoJSON helps evaluate satellite methods for detecting field burning. It contains 357 records; this selected cluster uses 125. [1]

Each row: field_id (unitless), field_name (text), field_category (unburnt / partial / complete), geometry (longitude and latitude in degrees). I derive field area in hectares from the polygons in UTM Zone 43N.

RESEARCH / CLOSEST PUBLISHED METHOD

Gupta et al. (2026) use a bivariate probit model for farmers' machinery adoption and burning decisions in Punjab. Time pressure, cost and access matter; partial and complete burning have different contexts. I borrow the idea of field-level decision probability, not their fitted coefficients. [2]

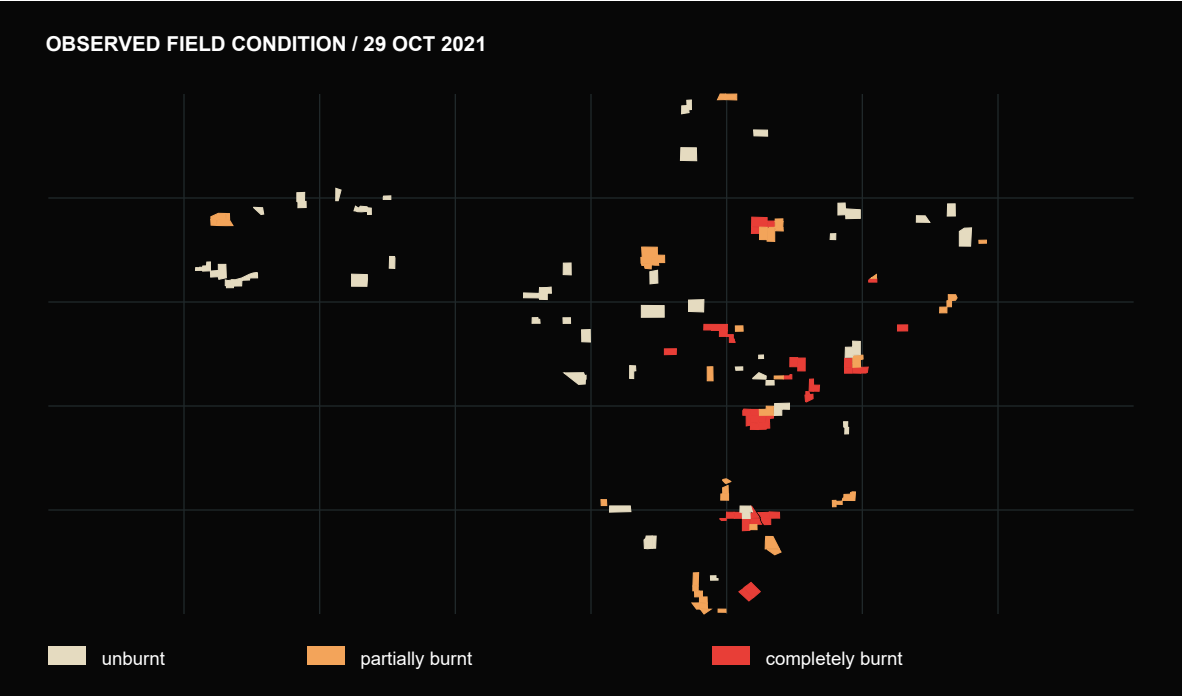


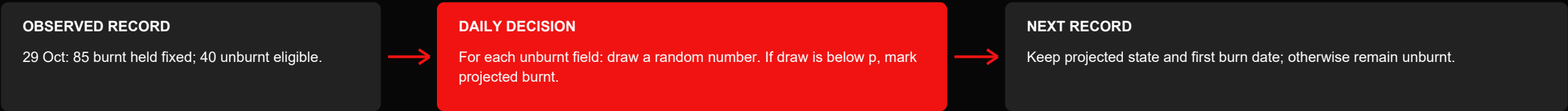
FIG. 01 The 125 selected CEEW field polygons, drawn from the Assignment 02 record.

[1] CEEW, Patterns of Crop Residue Burning (2026), doi:10.5281/zenodo.20179137. [2] Gupta et al. (2026), doi:10.1088/2976-601X/ae7a75.

FORTY FIELDS AT RISK

A proposed daily field-decision model for 30 October–12 November 2021. Observed partial and complete burns stay fixed; each of the 40 unburnt fields may change once. The diagram and rows below are a design for a future run, not calculated results.

THE RULE / ONE ROW, ONE DAY



PARAMETERS / ALL STARRED VALUES ARE MY ASSUMPTIONS

STATUS QUO *
p = 3% per eligible field per day

MORE CRM SUPPORT *
p = 1.5% per day (assumed 50% reduction)

RUN *
14 daily steps; 100 repeats per scenario

Chance: only the daily draw. All starred numbers are illustrative, especially p; Gupta et al. identify relevant constraints but provide no coefficient usable for these fields.

THE DATA IT WOULD GENERATE / EXAMPLE ROWS

ID	SCENARIO	RUN	DAY	STATE	FIRST BURN	AREA
312	status quo	1	0	unburnt	--	1.73 ha
312	status quo	1	5	proj. burnt	03 Nov	1.73 ha
313	status quo	1	14	unburnt	--	1.89 ha
314	CRM support	1	14	proj. burnt	08 Nov	1.53 ha

CANNOT SEE harvest timing / machinery access / real later burn dates / smoke and health effects

CHART SKETCH / SCENARIO COMPARISON

